INTELLIGENT SORTING SYSTEM

Step 1 : Choose the mechanical concept.

Step 2 : Make component inventory

Decide on voltage / current info

Step 3 : Communication b/w phone, ESP 32, laptop

Step 4 : Decide how the machine knows an item is present

Step 5 : Jam detection

Step 6 : CAD

Feeding mechanism:

Rotating disc + curved guide + gravity. chute ~~stopper~~.

COMPONENTS

A.) Feeding mechanism

- Rotating disc
- DC geared motor
- Motor mounting bracket
- circular disc
- curved outer guide.
- adjustable guide
- exit opening
- gravity chute
- chute side walls
- Base /frame
- bearings / bearings holder.

1 B) Inspection section

- Inspection platform
- Gate
- small servo
2 - side walls
- phone holder
3. - lighting
- background plate

C) Sorting mechanism

- servo motor
- flap / diverter
- servo mounting bracket
- sorting chute
- 3 collection bins
- bin mounting / frame

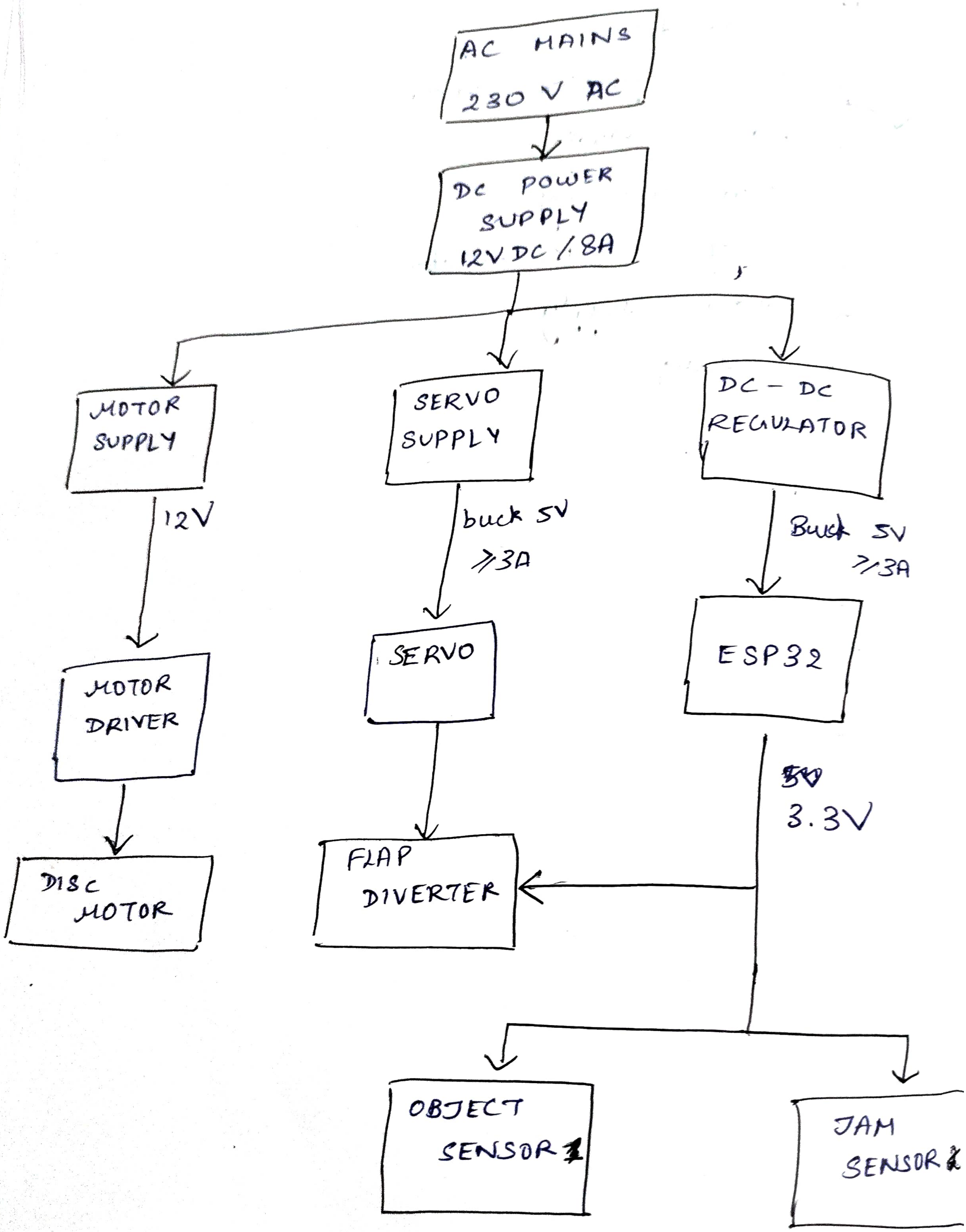
D) Electronics

- ESP 32
- DC motor driver
- DC geared motor
- 2 x servo motor
- IR obj sensor
- jumper wires
- connectors
- bread board
- switch

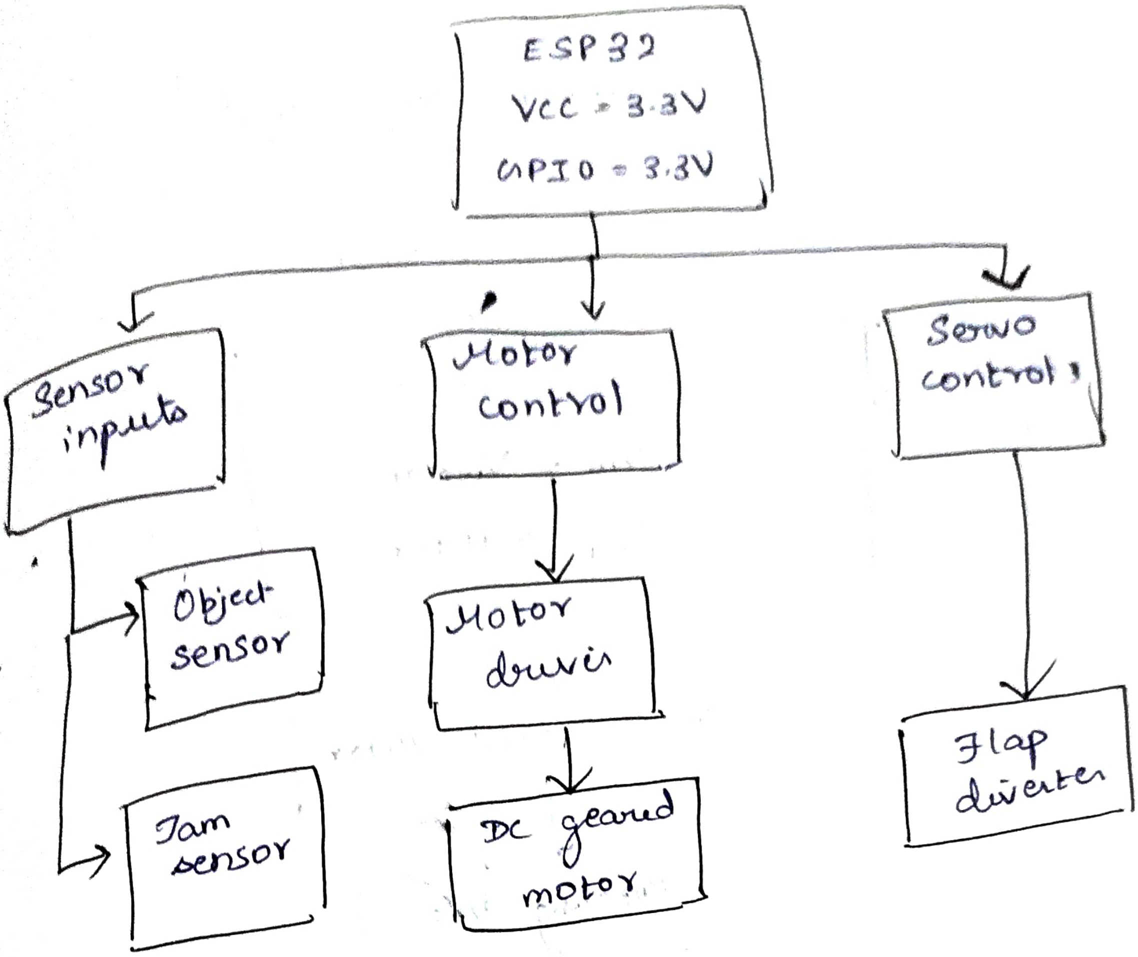
ELECTRONICS / CONTROL PART

1. Rotating disc motor
2. Motor driver
3. Object / position sensor
4. ESP32 controller
5. Jam recovery
6. Flap diverter actuator
7. Power supply
8. Wiring

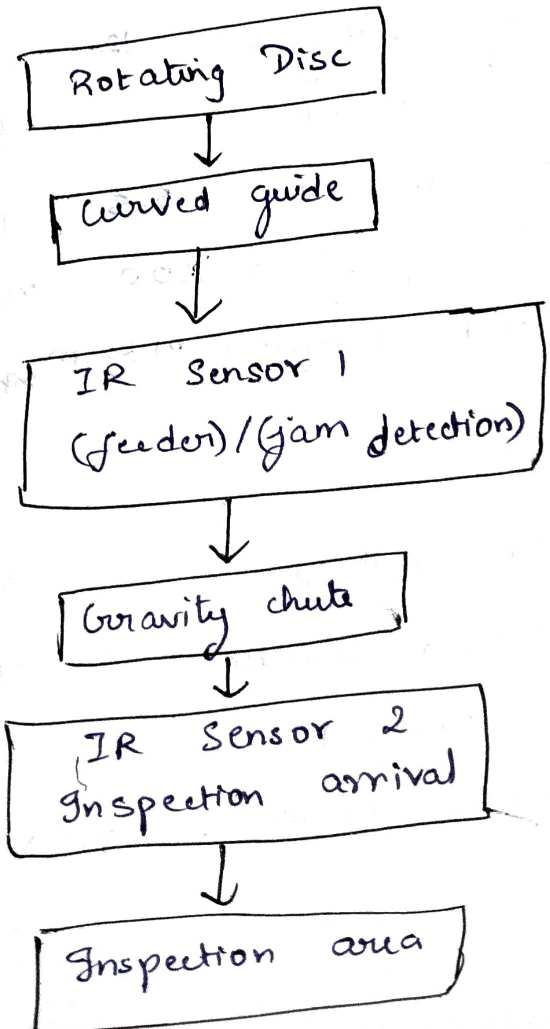
OVERALL ARCHITECTURE



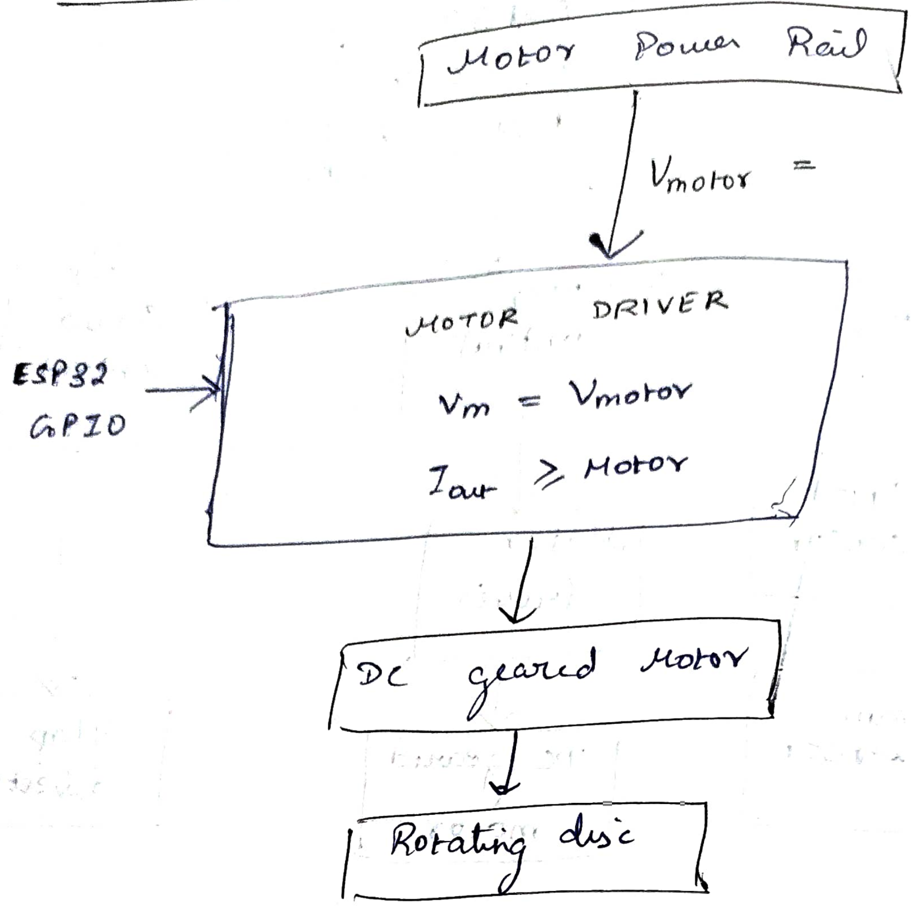
ESP32 ARCHITECTURE



SENSOR :



MOTOR CIRCUIT:



Motor

Operating voltage = 12 V

Rated current = A

Stall current = 5.5 A

Rated speed = 200 rpm

Required torque = 21 kg cm ~~Nm~~

Motor driver

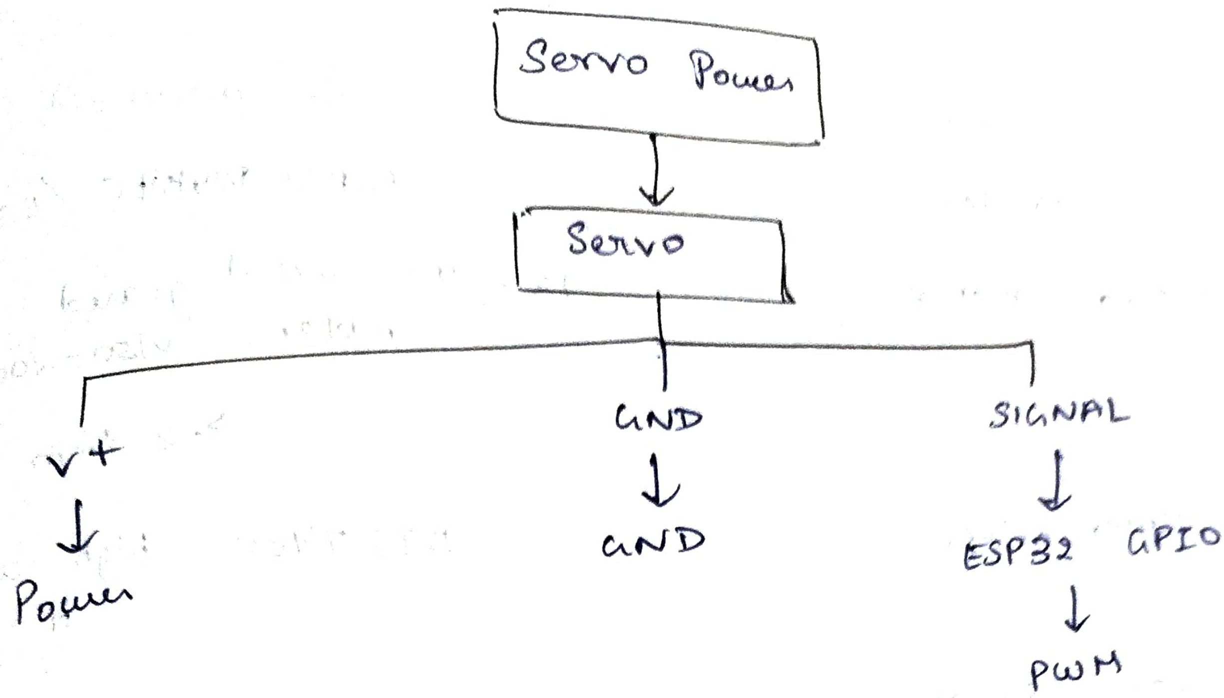
Motor voltage = V

Continuous current = A

Peak current = A

Logic voltage = 3.3V

SERVO / FLAP CIRCUIT :



Servo operating voltage : 5 V

Idle current : A

Operating current : A

Stall current : 1.8 A

Control interface : PWM

Signal voltage : 3.3V

COMMUNICATION

IR Sensors → ESP32 → Motor driver → Motor

↕ wifi

Laptop

↓

Phone (camera)

↓

YOLO

↓

ESP32

↓

Servo

IR Sensor → ESP32

OUT ⇝ GPIO

GND ⇝ GND [Digital GPIO]

VCC ⇝ 5V

ESP 32 → Servo

ESP32 GPIO ⇝ servo signal

5V supply ⇝ servo VCC [PWM]

GND ⇝ servo gnd

ESP32 → Motor driver

ESP32 GPIO ⇝ BTS 7960 control [Digital GPIO + PWM]

12V supply ⇝ BTS 7960 motor power

BTS 7960 ⇝ motor

ESP32 ↔ Laptop

wifi + MQTT

Laptop ↔ Phone

wifi video streaming